Sahoshi

Prototype evidence

Not by stopping. By steering.

One live picture of Kolkata's roads, rain and crowds for the traffic police, KMC, disaster response and ambulance (108) crews. On-premises open-source AI proposes; a named officer decides.

29 October 2026 · github.com/shubro18202758/sahoshi · MIT licence

1 · What exists today

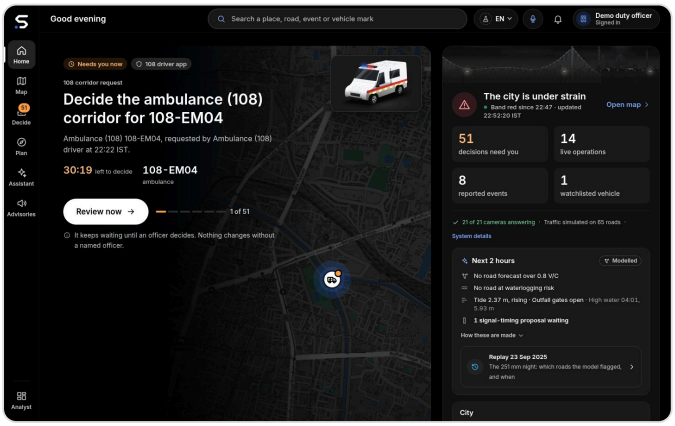
- **A working system on one on-premises AI computer** — the live stack reports READY 12/12: road store, live traffic ingest, decision engine, operator console (Bengali, English, Hindi), event emulator, OSINT ingest, three local language/embedding model hosts.
- **Five AI services** running beside it: corridor forecasting (Chronos-2), waterlogging nowcast (rain × Hooghly tide × terrain × pumps), a signal-timing traffic twin, Bengali voice (speech-to-text and read-aloud) and camera road state (Qwen3-VL).
- **Real Kolkata roads**: 49 junctions and 65 road segments (125 km), the Nabanna → airport route 29 km from OpenStreetMap; a 163-place gazetteer in English, Bengali and Hindi (wards, police stations, hospitals, pandals, ghats, low-lying spots).
- **Open feeds wired in**: IMD / NDMA SACHET alerts for West Bengal, Bengali and English Kolkata news, KMC notices and Kolkata Traffic Police advisories, Open-Meteo rainfall and air, NASA FIRMS fire hotspots, GDACS cyclones, CWC river gauges, USGS quakes, the Hooghly tide (modelled), Copernicus 30 m terrain, KMC FFEWS adapter (emulated).
- **Quality**: 69 automated gates green: 6,310 Python tests, 121 Rust tests, 46 console unit suites, contract replays and type checks.

Labels used throughout: **LIVE** (a real open feed), **OBSERVED** (a measured record), **MODELLED** (computed under stated assumptions), **SIMULATED** (an exercise scenario), **EMULATED** (a stand-in for a feed that needs an agency's access).

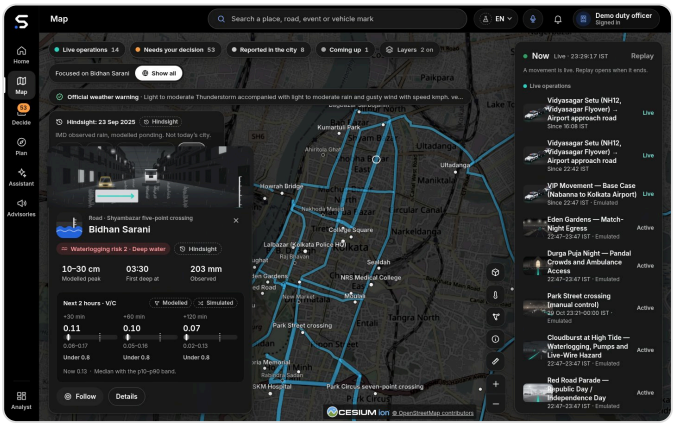
2 · The problem, in Kolkata

What	Fact	Source
Cloudburst at high tide	251.6 mm in 24 h on 22–23 Sep 2025, the heaviest in about four decades; roads went under within the hour; 8–12 deaths, most by electrocution; metro, rail and ~30 flights disrupted	IMD; Reuters/PTI via Dawn, Business Standard
Drainage and the river	Outfall gates close at high tide on the Hooghly — the same rain waterlogs far faster	KMC/ADB FFEWS project; port statements
Flood warning infrastructure	KMC is building India's first city-level Flood Forecasting & Early Warning System (ADB) — sensors in canals, pumping stations and junctions	ADB "Toward Resilient Kolkata"; Millennium Post
Durga Puja	10,000+ police a day, ~30% on traffic; goods-vehicle restrictions; pandal-route apps	Kolkata Police via Millennium Post (2025)
Bridges	Howrah Bridge: ~1 lakh vehicles and ~1.5 lakh pedestrians a day	Wikipedia; Deccan Herald (approximate)
Coordination	Traffic police, KMC, WBSDMA, the power utility and ambulance (108) each hold one piece; coordination by phone and radio	observed practice

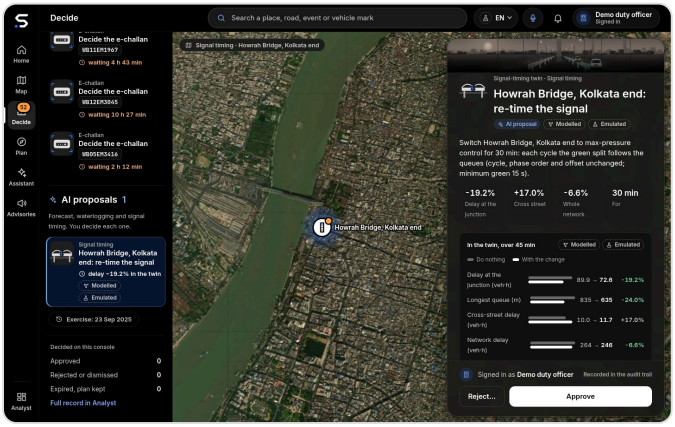
3 · The screens



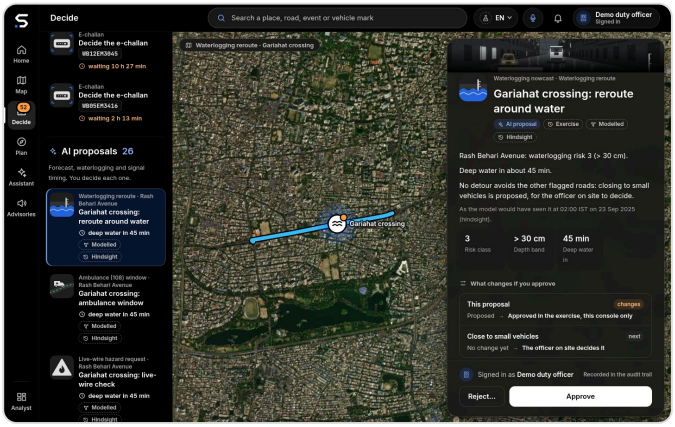
Home — what needs the officer now, Rain · Tide, the next two hours



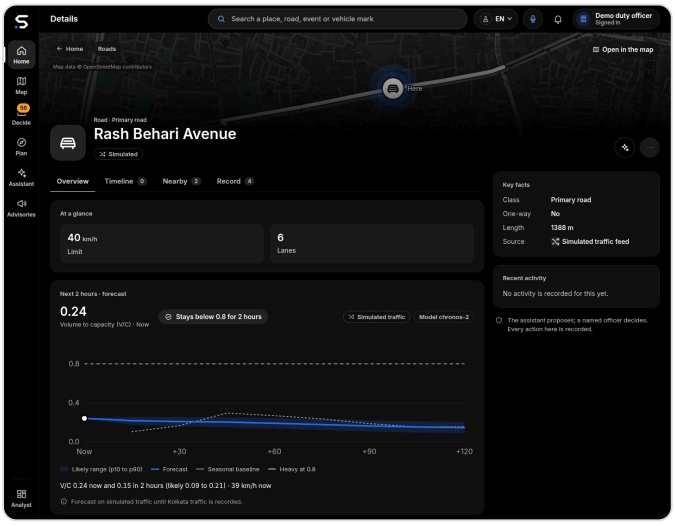
Map — waterlogging risk on every road, with its drivers labelled



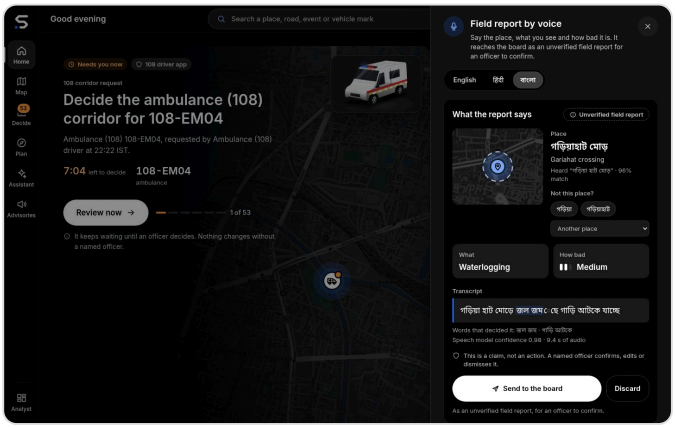
Decide — a signal-timing proposal tested in the traffic twin against doing nothing



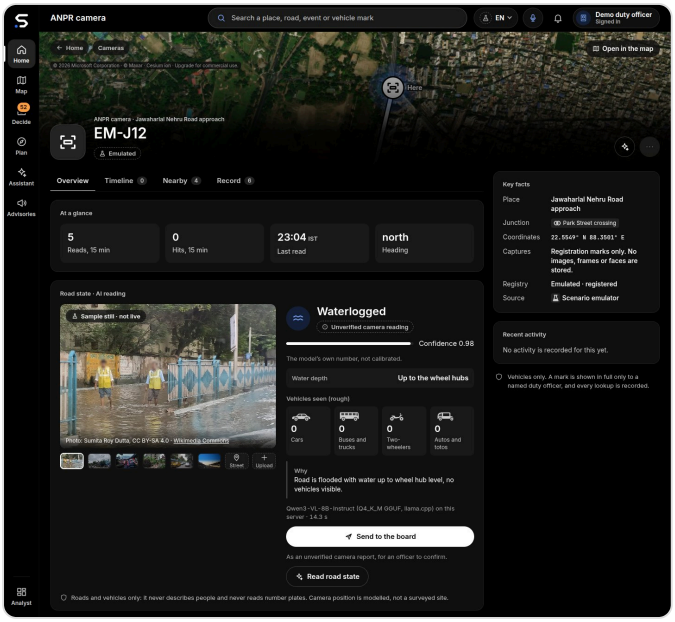
Decide — reroute, ambulance window, live-wire hazard request, Bengali advisory



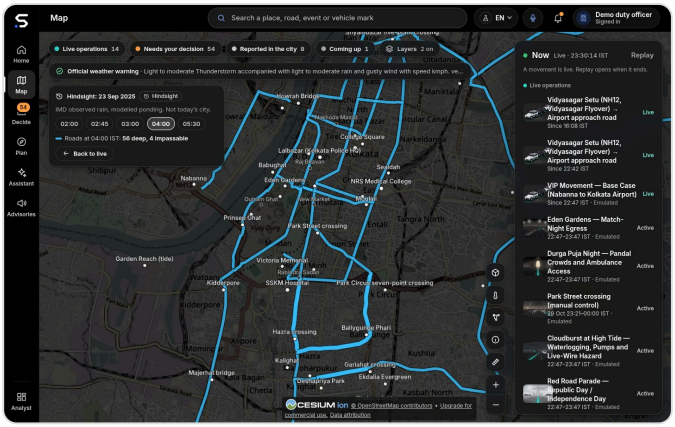
A road — two-hour forecast with its band, waterlogging class and drivers



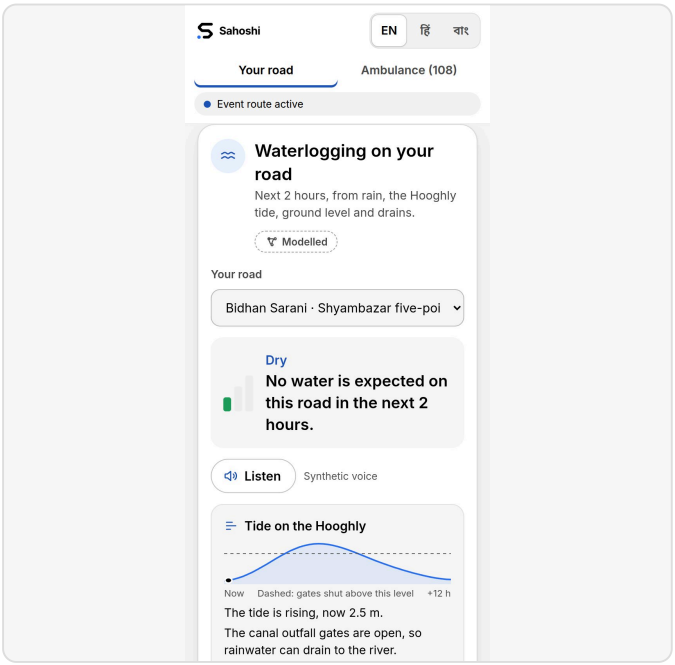
A field report spoken in Bengali becomes a structured, unverified report



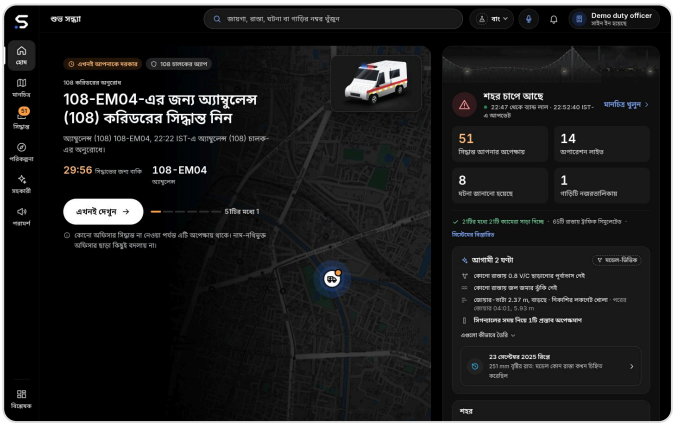
Camera road state — roads and vehicles only; person questions are refused



Replay of the 22–23 Sep 2025 storm (IMD-observed rain, modelled ponding)

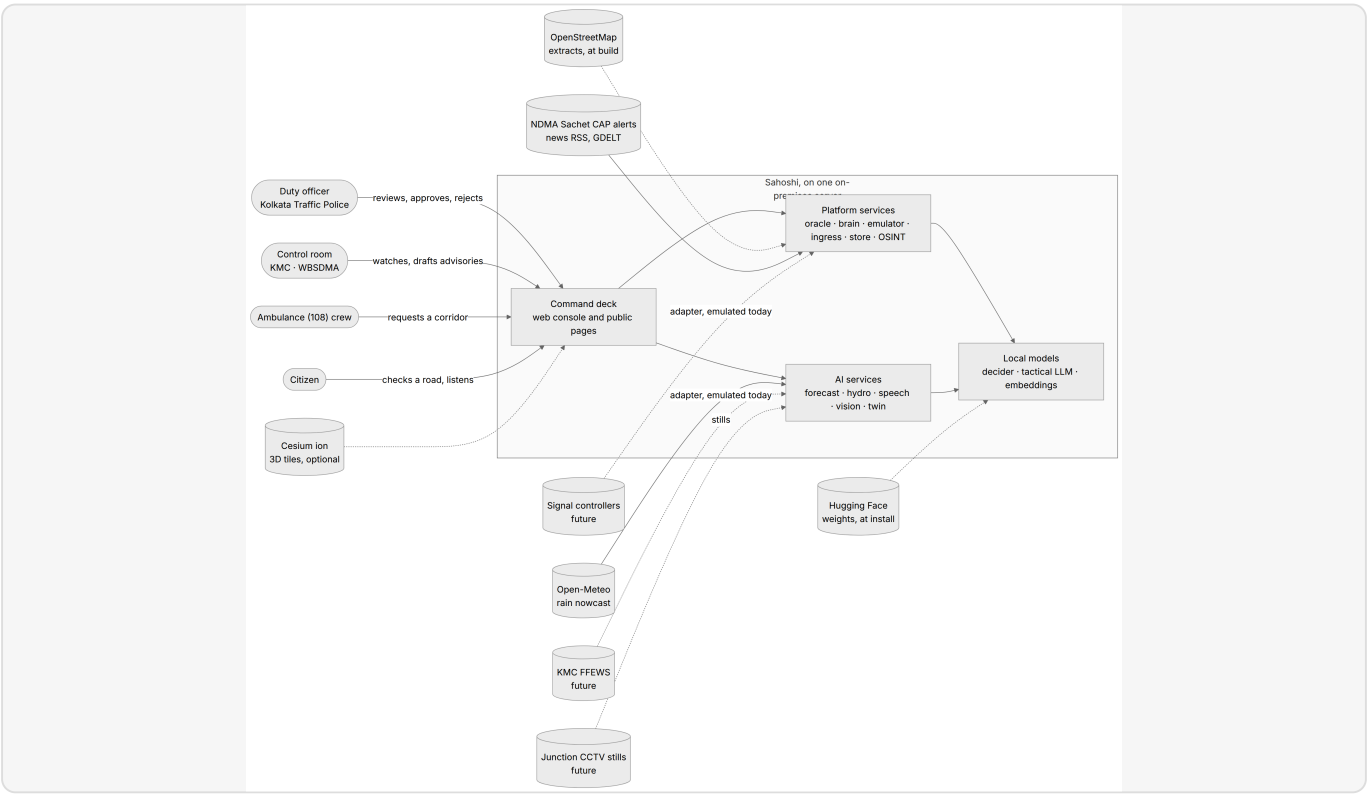


Citizen page — waterlogging on your road, with Listen

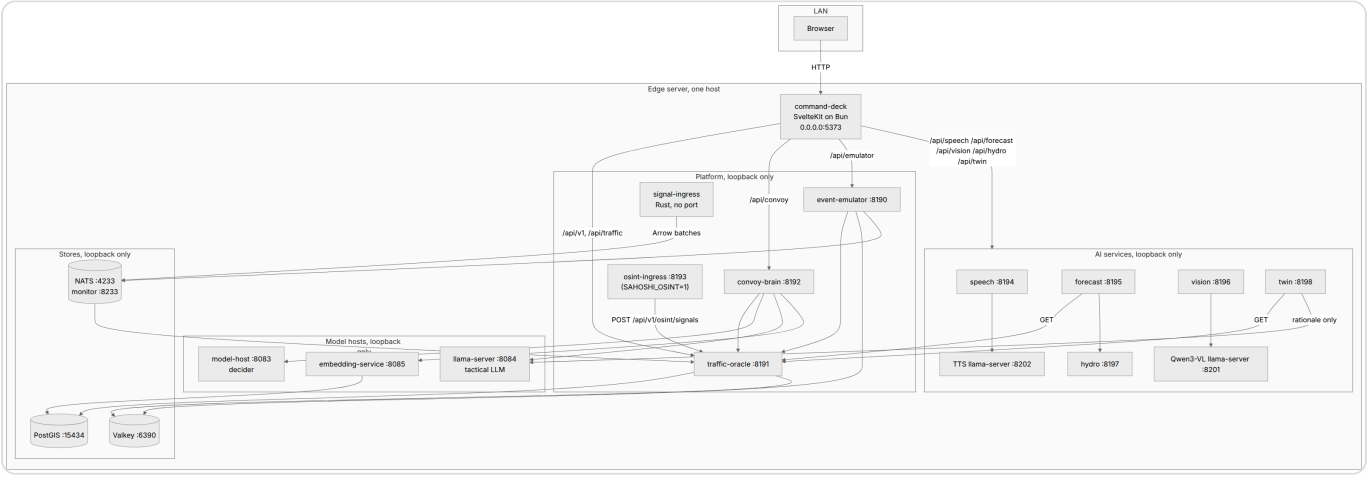


The console in Bengali

4 · Architecture



System context — everything runs on premises; the deck is the only network-facing port.



Services and ports.

5 · The AI, measured

Capability	Model (licence)	Result	Label
Corridor forecasting	Chronos-2 (Apache-2.0)	MAE 0.043 V/C vs 0.095 for the seasonal baseline (−55%); p10–p90 coverage 0.80	SIMULATED history until Kolkata traffic is recorded
Waterlogging nowcast	rain × tide × DEM × pumps	Replay of 22–23 Sep 2025 on IMD-observed rain: 13 of 13 reported on-network places flagged; first flag 02:45 IST	OBSERVED rain, MODELLED ponding
Signal-timing twin	cell-transmission model + Webster / max-pressure	Bottleneck discharge exact; shockwave speed within 0.2% of theory; 30 min of the city in 0.12 s	MODELLED
Bengali voice	Bengali Whisper-medium (Apache-2.0), AI4Bharat Indic Parler-TTS	Speech-to-text WER 11.2%, CER 2.5%, place 55/60, kind 60/60 (IndicConformer measured, not better); voice round trip WER 8.2%, places 60/60 (Svara 11.2%, 55/60)	synthetic round trip
Road state from a still	Qwen3-VL-8B (Apache-2.0)	11 of 12 correct; 9 of 9 person/plate probes refused; live on this computer in 2.4 s (Qwen3.5-9B measured: 10 of 12)	Commons photos, credited
Decision support	pplx-decider-v1.2 (Perplexity, Apache-2.0, 27B)	22 of 26 labelled Kolkata decisions; answers in 1.0 s at p95 (was 2.1 s with v1-27b)	proposals only; an officer decides
Memory and retrieval	pplx-embed-context-v1-4b + pplx-embed-v2-late-0.6b (MIT)	Bengali SOP recall@1 0.93 vs 0.71 for bge-m3; cross-script news dedup AUC 0.997 vs 0.56 (word overlap)	synthetic Kolkata eval set
Privacy (OSINT)	PII-Tracer (MIT) + rules	person recall 0.84 → 0.94; 0 of 122 place names redacted	synthetic eval set

Every model runs on the one on-premises computer; numbers come from the services' own evaluation files in the repository (docs/ai/*.md).

6 · Expected impact (MODELLED)

Measure	Value	Basis
Junction-freeze delay	100× less (range 25×–506×)	rolling ~2.5 min holds instead of a 25 min freeze (central case); delay scales with the square of the hold
VIP movement, Nabanna → airport	357,188 → 3,572 commuter-minutes	13 junctions, 29 km on the Kolkata network
Ambulance (108)	≈ 33,000 golden-hour minutes a year	12 crossing trips per movement-day (assumption)
Waterlogging warning	45–120 minutes before deep water	replay of 22–23 Sep 2025 on IMD-observed rain

Model outputs on Kolkata's OpenStreetMap network under stated assumptions (IMPACT_REPORT.md) — not field measurements, and never a claim about lives saved.

7 · Tests and quality

Suite	Result
traffic-oracle	2,106 passed
convoy-brain	800 passed
OSINT (common, ingress, geo, analysis)	2,032 passed
event-emulator	367 passed
embedding-service	191 passed
corridor-store	153 passed
speech · vision · twin · forecast · hydro · city feeds	219 · 102 · 50 · 46 · 26 · 113 passed
signal-ingress (Rust)	121 passed
command-deck	45 unit suites, contract replays, svelte-check 0 errors, design-token and copy lints (4,908 strings)
Total	69 of 69 gates green · 6,310 Python + 121 Rust tests

8 · Implementation and cost

When	What
Month 0–1	Read-only data MoU: Kolkata Traffic Police (CCTV stills, signal plans), KMC (FFEWS), WBSDMA (alerts), the ambulance (108) operator; nodal officers.
Month 1–2	Install the on-premises unit in the pilot control room; switch adapters from EMULATED to LIVE as access arrives.
Month 2–4	Shadow mode — Sahoshi proposes, officers work as usual, weekly comparison.
Month 5–12	Live pilot in one division through the monsoon and Durga Puja; officers approve every action.
Month 12	Independent review; city-wide plan.

Cost item	Pilot (12 months)
On-premises AI units (2, one as backup)	₹8–10 lakh
Engineering (3 × 6 months + 1 × 6 months)	₹29 lakh
Water-level sensors where FFEWS has none (optional)	₹4 lakh
Training and SOP workshops	₹2 lakh
Backup connectivity	₹0.5 lakh
Contingency (15%)	≈ ₹6–7 lakh
Total pilot	**≈ ₹45–55 lakh**
City-wide, two years	≈ ₹2–2.5 crore; running ≈ ₹40–60 lakh a year
Software licences	₹0 — open models, OpenStreetMap, Sahoshi (MIT)

9 · Scalability — a city is a folder

Everything city-specific lives in a City Pack (`cities/<id>/`): the OpenStreetMap extract and corridors, gazetteer, agencies, emergency numbers, plate format, languages, scenarios and impact parameters. The prototype was first built for another Indian city; the same build tools now produce Kolkata, and a test builds both. Next packs: Howrah and Bidhannagar (inside the Kolkata pack), Siliguri, Asansol–Durgapur, Patna and Ranchi (Hindi is already first-class).

10 · Guardrails

- Vehicles only — no faces, no persons; number plates masked until a named duty officer is on duty.
- The AI proposes; a named officer approves every action; no answer — the current plan holds; every decision audited.
- Honest labels on every number and every feed not yet connected.
- Events, not people: news and social signals are aggregate and about places; posters are never stored or scored.
- No e-challan can be approved against a theft victim.
- On-premises, retention limits, in the spirit of the DPDP Act 2023.

Sources

1. Reuters / PTI on the 22–23 Sep 2025 Kolkata rain — dawn.com/news/1944310; business-standard.com (2025-09-23)
2. ADB, "Toward Resilient Kolkata: India's first comprehensive city-level flood forecasting and early warning system" — adb.org/publications/toward-resilient-kolkata
3. Millennium Post — KMC unveils city-level flood forecasting warning system; Kolkata Police Puja Bandhu app; 10K police for Puja duty (2025)
4. Howrah Bridge — en.wikipedia.org/wiki/Howrah_Bridge; Deccan Herald
5. Census of India 2011 — Kolkata Municipal Corporation and urban agglomeration population
6. Model cards: amazon/chronos-2, google/timesfm-2.5, Qwen/Qwen3-VL-8B-Instruct, perplexity-ai/pplx-embed, bengaliAI Whisper-medium, kenpath/svara-tts-v1 (Hugging Face)
7. OpenStreetMap contributors (ODbL 1.0); Copernicus DEM GLO-30; Open-Meteo; NDMA SACHET; GDACS; USGS